Table 1 Tools for designing DNA sequence libraries and for related sequence-design problems. Rows appear in the same order as the columns of Table 2. The final two rows group tools that address problems adjacent to library design; their members are named in the table and cited individually. Tools were examined in August 2026.

Tool	Purpose	Key features	Output	Availability	Ref.
DNA Chisel	Optimize a single sequence against declared constraints and objectives	Hard constraints and scored objectives combined in one problem; codon optimization and avoidance of restriction sites, homopolymers and extreme GC	One optimized sequence with a constraint-satisfaction report	Python package (PyPI); web application	[1]
MPRA Design Tools	Design MPRA libraries and estimate statistical power	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates	R package (GitHub); Shiny application	[2]
MPRAnator	Design MPRA libraries of motif arrangements and sequence variants	Combinatorial placement of motifs; randomly sampled and combinatorial substitution sets; scrambled and reverse-complement controls	Library of sequences (FASTA)	Web service	[3]
Mutation Maker	Design mutagenic primers for laboratory construction of variant libraries	Three workflows: site-scanning saturation mutagenesis, multi-site directed mutagenesis, and PCR-based gene synthesis; degenerate codons and primer mixing ratios	Mutagenic primer pairs, or gene fragments with mixing ratios	Web application (Docker); REST API	[4]
Oligopool Calculator	Design library infrastructure and analyze sequencing readout	Constraint-aware barcodes, primers and spacers; degenerate compression of a supplied variant set; off-target screening against a host genome	Synthesis-ready library; variant counts from sequencing reads	Python package (PyPI); command line	[5]
PoolParty	Design DNA sequence libraries of any type	Libraries specified as a directed acyclic graph of operations; saturation mutagenesis, randomly sampled variants and pairwise and higher-order variants can be mixed in one library; per-variant design cards	Library of sequences, each with a design card (CSV, TSV, FASTA, JSONL)	Python package (PyPI)	This work
tangermeme	Probe cis-regulatory logic with deep learning models	Saturation mutagenesis of input sequences; insertion and removal of motifs across sets of sequences	Perturbed sequences and model predictions	Python package (PyPI)	[6]
VaLiAnT	Design and annotate libraries for saturation genome editing and cDNA deep mutational scanning	Saturation mutagenesis from genomic coordinates; several mutation types per target region; transcript-aware, including codons split across exons	Per-region libraries with metadata tables and VCF carrying variant consequence annotation	Command-line tool (source, Docker)	[7]
General-purpose toolkits *(Biopython, pydna, SeqPro)*	Manipulate sequences, simulate cloning, prepare arrays for modeling	Parsing, transformation and file-format support; no library abstraction	Transformed sequences	Python packages (PyPI)	[8–10]
Single-sequence design tools *(CodonGenie, ledidi)*	Choose codons, or edit one sequence toward a model's objective	Degenerate codons covering a target residue set; gradient-based editing guided by a predictive model	A codon set, or one edited sequence	Web service; Python package	[11, 12]

Table 2 Capabilities of tools for designing DNA sequence libraries. ● denotes a capability the tool provides; ◐ one provided with restrictions; ○ one the tool does not provide. Capabilities are counted only where the tool provides a dedicated operation, parameter, or mode. Tools were assessed in August 2026.

	DNA Chisel	MPRA Design Tools	MPRAnator	Mutation Maker	Oligopool Calculator	**PoolParty**	tangermeme	VaLiAnT
Library Specification								
Library object	○	◐	○	◐	◐	●	◐	○
Composable operations	◐	○	◐	○	●	●	◐	○
Mixed variant types in one library	○	●	●	◐	○	●	○	●
Variant Generation								
Saturation mutagenesis	○	○	○	○	◐	●	●	●
Randomly sampled variants	●	◐	●	○	○	●	○	○
Pairwise and higher-order variants	●	○	●	●	◐	●	◐	○
Model-guided variants	●	○	○	○	○	◐	●	○
Variant Types								
Codon / amino-acid substitutions	●	○	○	◐	○	●	○	●
Insertions and deletions	○	●	◐	○	○	●	●	●
Combinatorial multi-motif placement	○	○	●	○	○	●	●	○
Recombination	○	○	○	○	○	●	○	○
Shuffling	○	○	◐	○	○	●	●	○
Physical Construction								
Synthesis-constraint checking	●	◐	◐	○	●	◐	○	◐
Constraint-based optimization	●	◐	◐	●	●	○	○	○
Primer design	◐	○	○	●	●	○	○	○
Metadata and Inspection								
Per-sequence design cards	●	●	◐	●	●	●	○	●
Automatic naming	○	◐	●	●	◐	●	◐	●
Sequence styling	○	○	◐	○	○	●	○	○
Genomic Integration								
Genome coordinates	○	◐	●	○	○	◐	◐	●
Transcript / annotation aware	○	○	○	○	○	○	○	●

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